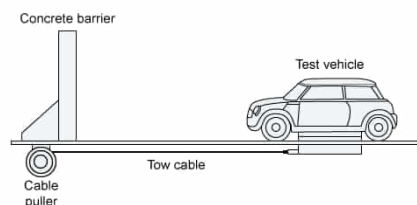
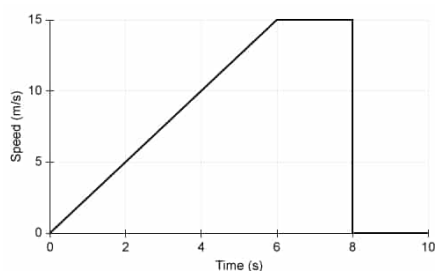


Vehicle crash testing is used to study the physics of passenger injuries and vehicle safety systems. Crash testing can simulate head-on collisions, impacts to the side of a vehicle, rollover crashes, and other situations. Likewise, crash testing can evaluate the efficacy of a range of safety devices, such as seat belts, airbags, and automatic braking systems. An automatic braking system uses sensors to detect stationary obstructions in front of the car. Based on the vehicle's speed and the distance to the object, the system automatically engages the car's brakes to avoid the collision or at least reduce the potential for injuries.

In a crash test, a vehicle is pulled toward a concrete barrier using a steel tow cable, as shown in Figure 1. The barrier is designed so most of the energy of the collision is transferred to the vehicle. Figure 2 shows the vehicle speed over time as it is accelerated to the desired speed and strikes the barrier.



**Figure 1** Schematic of a crash test setup



**Figure 2** Vehicle speed during a crash test

Crash test dummies are used to determine which injuries could occur to the vehicle's passengers. Various sensors within the dummies record important data, such as body acceleration, deformation of the chest cavity, and forces applied to limb bones. A dummy may contain 50 different sensors and collect 30,000 individual data points during a crash lasting 0.1 s.

A 1,200 kg car is traveling at 10 m/s. What is the minimum distance at which an automatic braking system needs to activate to avoid hitting a stationary object? (Note: The braking system can decelerate the car at  $2 \text{ m/s}^2$ .)

- ☐ A. 2.5 m [8%]
- ☐ B. 5.0 m [22%]
- ✓ ☒ C. 25 m [59%]
- ☐ D. 50 m [10%]

Correct

58% answered correctly

**Explanation:**

**Stopping distance of a car**

$$v^2 = v_0^2 + 2a \cdot \Delta x$$

$$\Delta x = \frac{v^2 - v_0^2}{2a}$$

- $v$  Final velocity
- $v_0$  Initial velocity
- $a$  Acceleration
- $\Delta x$  Displacement

$$\Delta x = \frac{\left(0 \frac{\text{m}}{\text{s}}\right)^2 - \left(10 \frac{\text{m}}{\text{s}}\right)^2}{2 \left(-2 \frac{\text{m}}{\text{s}^2}\right)} = \frac{-100 \frac{\text{m}^2}{\text{s}^2}}{-4 \frac{\text{m}}{\text{s}^2}} = 25 \text{ m}$$

$\Delta x = 25 \text{ m}$

The **translational motion equations** relate kinematic parameters such as **acceleration  $a$** , **displacement  $\Delta x$** , **velocity**, and time. Specifically, the square of the final velocity  $v$  of an object is equal to the sum of the square of the initial velocity  $v_0$  and twice the product of  $a$  and  $\Delta x$ :

$$v^2 = v_0^2 + 2a\Delta x$$

Rearranging the above equation to solve for  $\Delta x$  yields:

$$\Delta x = \frac{v^2 - v_0^2}{2a}$$

In this question,  $v = 0$  m/s,  $v_0 = 10$  m/s, and  $a = -2$  m/s<sup>2</sup> because the car is decelerating. Substituting these values into the equation above gives:

$$\begin{aligned}\Delta x &= \frac{\left(0 \frac{\text{m}}{\text{s}}\right)^2 - \left(10 \frac{\text{m}}{\text{s}}\right)^2}{-4 \frac{\text{m}}{\text{s}^2}} \\ \Delta x &= \frac{-100 \frac{\text{m}^2}{\text{s}^2}}{-4 \frac{\text{m}}{\text{s}^2}} = 25 \text{ m}\end{aligned}$$

Therefore, the automatic braking system should activate when the vehicle is 25 m away from the object.

**(Choice A)** 2.5 m results from not squaring the value for  $v_0$ .

**(Choice B)** 5.0 m is obtained if the translational motion equation that includes time,  $v = v_0 + at$ , is used and the value of  $t$  is mistaken for the value of  $\Delta x$ .

**(Choice D)** 50 m is calculated if the acceleration is not multiplied by 2.

**Educational objective:**

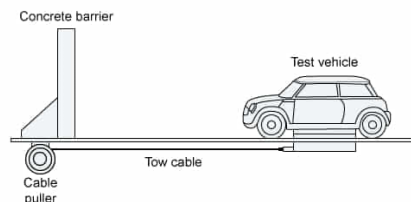
The translational motion equation can be used to determine an object's displacement during constant acceleration when the initial and final velocities are known. For a vehicle, the stopping distance is calculated by setting the final velocity to zero.

**Time spent:** 0 sec  
**Subject:**  
Physics

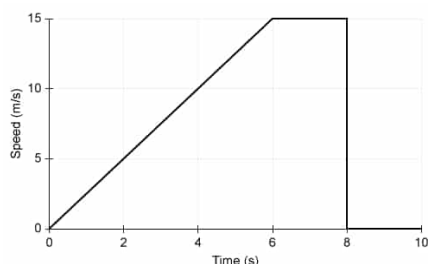
**QID:** 403589  
**System:**  
Mechanics and Energy

Vehicle crash testing is used to study the physics of passenger injuries and vehicle safety systems. Crash testing can simulate head-on collisions, impacts to the side of a vehicle, rollover crashes, and other situations. Likewise, crash testing can evaluate the efficacy of a range of safety devices, such as seat belts, airbags, and automatic braking systems. An automatic braking system uses sensors to detect stationary obstructions in front of the car. Based on the vehicle's speed and the distance to the object, the system automatically engages the car's brakes to avoid the collision or at least reduce the potential for injuries.

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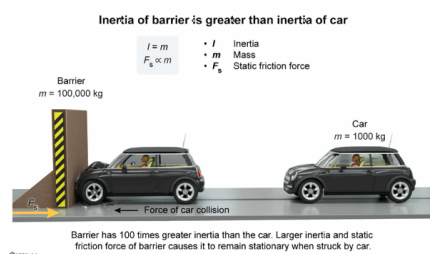
Researchers collide a 1,000 kg car into a stationary concrete barrier with a mass of 100,000 kg. Which of the following statements best explains why the barrier remains stationary when it is struck by the car?

- ☐ A. The coefficient of static friction for the barrier is greater than the coefficient of kinetic friction for the car. [29%]
- ☐ B. The kinetic energy of the car is greater than the kinetic energy of the barrier. [2%]
- ☐ C. The normal force the ground exerts on the car is greater than the normal force the ground exerts on the barrier. [2%]
- ☒ D. The inertia of the barrier is much greater than the inertia of the car. [64%]

Correct

64% answered correctly

### Explanation:



According to [Newton's first law of motion](#), an object in motion stays in motion and an object at rest stays at rest. An object's ability to resist changes in speed depends on its **inertia**  $I$ , which equals its **mass**  $m$ :

$$I = m$$

Furthermore, the force required to move a stationary object must be greater than the maximum static friction force  $F_s$ , which is proportional to  $I$  because the normal force  $F_N$  is proportional to  $m$ :

$$F_s = \mu_s F_N = \mu_s mg$$

$$F_s \propto m$$

$$I \propto F_s$$

In this question, the barrier's inertia is 100 times greater than the car's inertia due to their difference in mass. Hence, the force required to overcome  $F_s$  and move the barrier is very large because of the barrier's much larger  $I$ . The force of the car on the barrier during the collision is smaller than  $F_s$ , and therefore the barrier remains stationary when the car collides into it.

**(Choice A)** The coefficients of static and kinetic friction could influence the motion of objects during a collision. However, the friction force also depends on the object's normal force, which is much greater for the barrier due to its greater mass. Therefore, the barrier's inertia has a greater influence on whether it remains stationary than the difference in coefficients of friction.

**(Choice B)** The kinetic energy of an object is not related to its ability to resist changes in its speed.

**(Choice C)** The normal force exerted on an object by a surface is proportional to its mass. Therefore, the car's normal force is less than the barrier's normal force because the car's mass is 1/100th the barrier's mass.

**Educational objective:**

Inertia refers to the ability of an object in motion to stay in motion and an object at rest to stay at rest. An object's inertia depends on its mass and is also proportional to the maximum static friction force it experiences when stationary. Therefore, a stationary object with a large inertia could remain stationary when another smaller object strikes it.

Time spent: 0 sec

Subject:

Physics

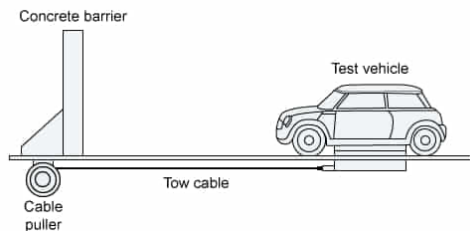
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System:

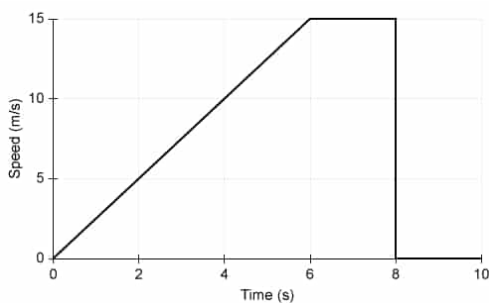
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A steel tow cable is used to pull a car at a constant velocity toward a barrier during a crash test. What information about the test is needed to calculate the tension in the cable?

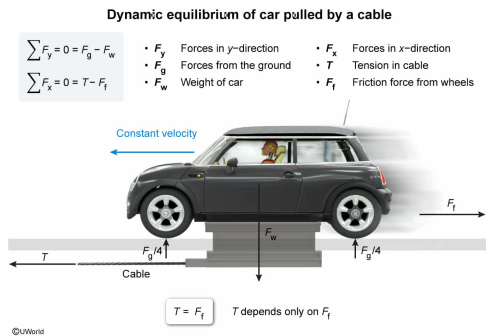
- I. Friction force on the car
- II. Mass of the car
- III. Velocity of the car

- ✓ ☒ A. I only [44%]  
☐ B. II only [34%]  
☐ C. III only [1%]  
☐ D. II and III only [20%]

Correct

43% answered correctly

## Explanation:



**Mechanical equilibrium** occurs when the net external force acting on an object is equal to zero. This does not imply that no forces are acting on the object, but rather that all forces acting on the object are balanced (ie, **zero net force** is exerted on the object). Forces are often separated into their **components** in two orthogonal directions (eg, x-axis forces  $F_x$  and y-axis forces  $F_y$ ) or three directions if considering three-dimensional space. Therefore, equilibrium occurs when these forces sum to zero:

$$\sum F_y = 0$$

$$\sum F_x = 0$$

There are two forms of mechanical equilibrium. **Static equilibrium** occurs when the object has zero velocity, and **dynamic equilibrium** occurs when the object has a *constant nonzero velocity*.

In this question, the vertical forces acting on the vehicle (y-direction) are the **weight**  $F_w$  of the vehicle and the force from the ground  $F_g$  acting on the tires.  $F_g$  is spread over four tires, so each tire experiences approximately  $\frac{F_g}{4}$ :

$$\sum F_y = \frac{F_g}{4} + \frac{F_g}{4} + \frac{F_g}{4} + \frac{F_g}{4} - F_w = F_g - F_w = 0$$

Similarly, the horizontal forces (x-direction) are the **tension**  $T$  on the steel tow cable and the **friction force**  $F_f$  on the car:

$$\sum F_x = T - F_f = 0$$

Solving the above equation for  $T$  gives:

$$T = F_f$$

Therefore,  $T$  depends only on  $F_f$  when the vehicle is being pulled at a constant velocity (**Number I**).

**(Number II)** The car's mass does affect the friction force on the car, and therefore the cable tension. However, the friction force also depends on the coefficient of friction and cannot be calculated only from the car's mass.

**(Number III)** The car's velocity does not affect the cable tension. The cable tension equals the friction force on the car, which depends only on the car's normal force and the coefficient of friction.

### Educational objective:

Dynamic equilibrium occurs when all external forces acting on an object sum to zero and the object moves with constant velocity. When pulling a vehicle at constant velocity, the magnitude of the pulling force equals the friction force on the car but acts in the opposite direction.